

Amazon ECS Cluster: Add EC2 Later After Cluster Creation

To Begin with the Lab

Summary of the Lab

In this lab, an **Amazon ECS cluster** was created without adding EC2 during the initial cluster setup. Later, an **ECS-optimized EC2 instance** was launched and manually added to the cluster. The process included selecting the correct ECS AMI, attaching the **ecsInstanceRole**, configuring the ECS cluster name in `/etc/ecs/ecs.config`, and restarting the ECS service so the instance could register itself with ECS. After that, the instance appeared in the cluster infrastructure tab as a container instance. The video also explained that this approach is useful when you first choose Fargate but later decide to add EC2 capacity.

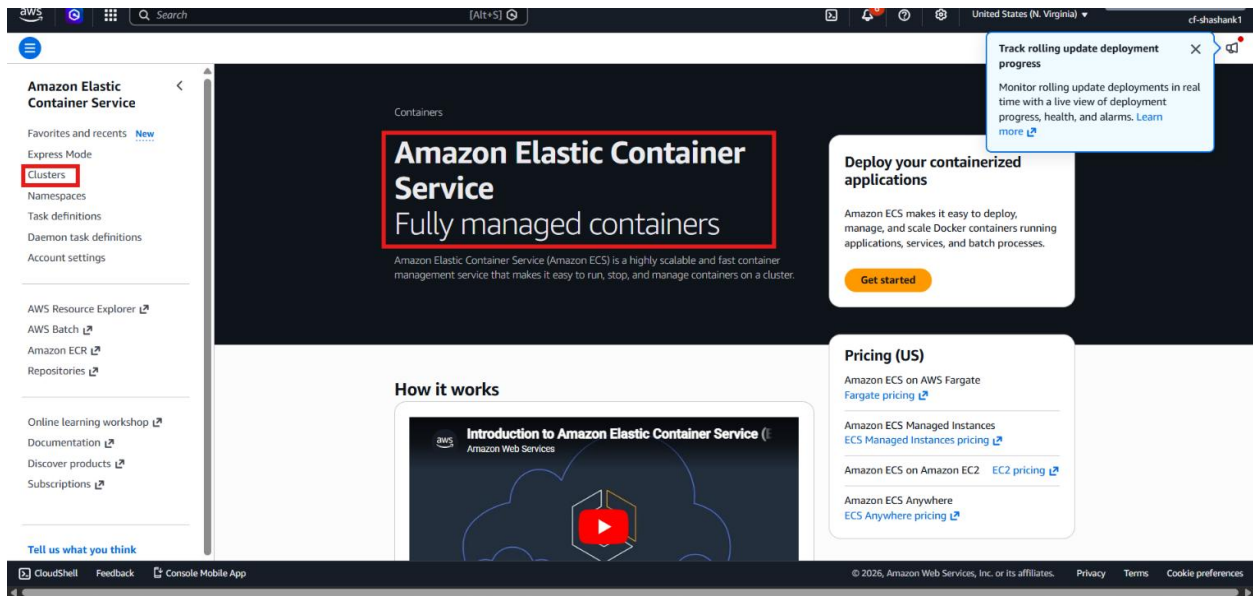
Understanding EC2 Addition After ECS Cluster Creation

When you create an ECS cluster without EC2, you can still add EC2 later. This is useful when you are not sure at the beginning whether you need serverless containers or EC2-based containers. The transcript explained two later-addition choices: launching an individual EC2 instance or creating an Auto Scaling Group manually. This video focused on the first option: launching one EC2 instance and registering it with the ECS cluster. Using an **ECS-optimized AMI** makes the setup easier because the ECS agent and container runtime are already installed.

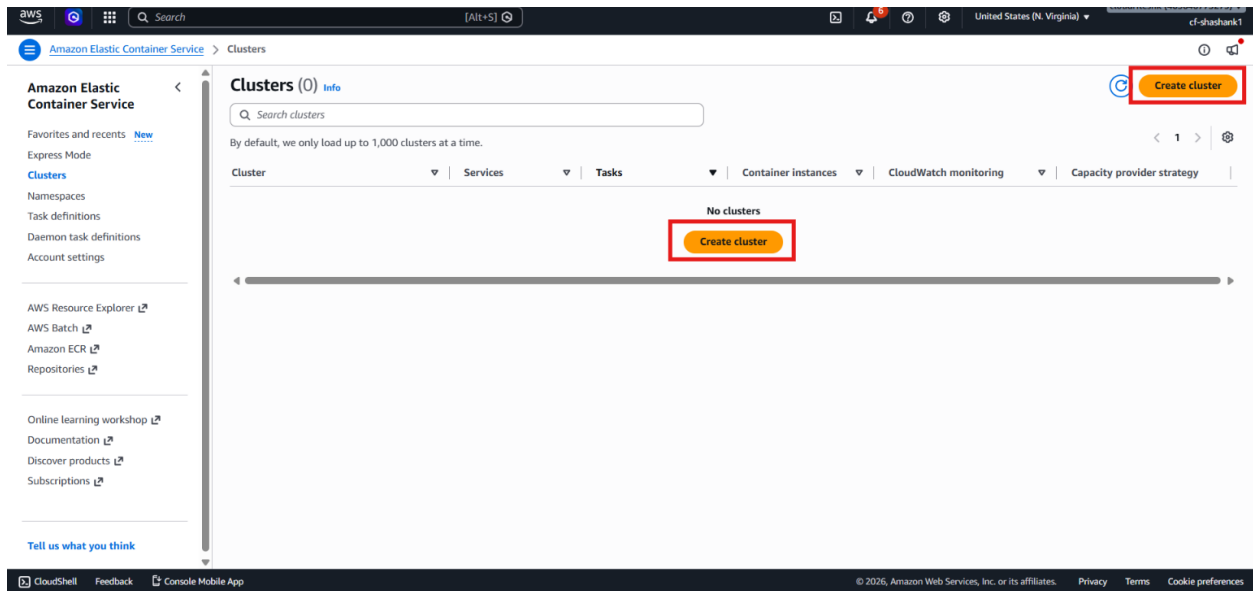
Lab Steps

Step 1 – Open ECS and Create a Cluster

- Sign in to the AWS Management Console.
- Open **Amazon ECS**.
- Go to **Clusters**.



- Click **Create cluster**.



- Enter the cluster name: ClusterWithoutFargate
- Do not add extra infrastructure for this first test.
- Keep Fargate selected.
- Create the cluster.

Create cluster Info

An Amazon ECS cluster groups together tasks, and services, and allows for shared capacity and common configurations. All of your tasks, services, and capacity must belong to a cluster.

Cluster configuration

Cluster name

ClusterWithoutFargate

Cluster name must be 1 to 255 characters. Valid characters are a-z, A-Z, 0-9, hyphens (-), and underscores (_).

► **Service Connect defaults - optional**

▼ **Infrastructure - advanced** Info

Configure the manner of obtaining compute resources that will be used to host your application.

Select a method of obtaining compute capacity

Your cluster is automatically configured for AWS Fargate (serverless), but you may choose to add Amazon EC2 instances (servers).

☒ **Fargate only**

Serverless instances - you don't think about creating or managing servers. Great for most common workloads.

☐ **Fargate and Managed Instances**

Managed instances - Amazon ECS will manage patching and scaling on your behalf while giving you configurability about the types of instances. Great for more advanced workloads.

☐ **Fargate and Self-managed instances**

Self-managed instances - you must ensure the instances are patched and scaled properly, and you have full control over the instances.

► **Monitoring - optional**

Configure observability, encryption, and logging options to maintain compliance and operational visibility of your container environment.

► **Encryption - optional**

- Observe that **Fargate** and **Fargate Spot** are still included automatically.

ClusterWithoutFargate

Last updated July 6, 2026, 13:03 (UTC+5:30)

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Cluster overview

Cluster ARN: [arn:aws:ecs:us-east-1:123456789012:cluster/ClusterWithoutFargate](#)

Status: Active

CloudWatch monitoring: Default

Registered container instances: 0

Services: 0 active | 0 draining

Tasks: 0 running | 0 pending

Capacity providers (2) Info

Last updated July 6, 2026, 13:03 (UTC+5:30)

[Manage tags](#) [Update](#) [Delete](#) [Create](#)

Filter capacity providers by property or value

Capacity provider	Scaling type	Provisioning model	Update status	Update status reason
Fargate	Fargate	-	-	-
Fargate Spot	Fargate Spot	-	-	-

Container instances (0) Info

Last updated July 6, 2026, 13:03 (UTC+5:30)

[Register external instances](#) [Actions](#)

Filter container instances by property or value

Container instance	Instance type	Status	Statu...	Type	Instance ID	Capacit...	Availability zo...	Running
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Step 2 – Launch an ECS-Optimized EC2 Instance

- Open Amazon EC2.
- Click **Launch instance**.

EC2

Dashboard
AWS Global View
Events

▼ **Instances**
Instances
Instance Types
Launch Templates
Spot Requests
Savings Plans
Reserved Instances
Dedicated Hosts
Capacity Reservations
Capacity Manager

▼ **Images**
AMIs
AMI Catalog

▼ **Elastic Block Store**
Volumes
Snapshots
Lifecycle Manager

▼ **Network & Security**

Resources
You are using the following Amazon EC2 resources in the United States (N. Virginia) Region:

Instances (running)	1	Auto Scaling Groups	0	Capacity Reservations	0
Dedicated Hosts	0	Elastic IPs	0	Instances	2
Key pairs	31	Load balancers	0	Placement groups	0
Security groups	29	Snapshots	0	Volumes	1

Launch instance
To get started, launch an Amazon EC2 Instance, which is a virtual server in the cloud.

Note: Your instances will launch in the United States (N. Virginia) Region

Launch instance **Migrate a server**

Instance alarms **View in CloudWatch**
0 in alarm 0 OK 0 insufficient data

Instances in alarm

Service health **AWS Health Dashboard**
Region: United States (N. Virginia)
Status: This service is operating normally.

Zones

Zone name	Zone ID
us-east-1a	use1-az4
us-east-1b	use1-az6
us-east-1c	use1-az1

EC2 cost
Date range: Past 6 months | Region: Global
Total cost: \$0.00
Regions: 6

Cost (\$)

0.0001
0.00008
0.00006
0.00004
0.00002
0

Jan 2026 Mar 2026 May 2026 Jul 2026

Date (UTC)

• Choose an ECS-optimized Amazon Linux AMI.

EC2 > **Instances** > Launch an instance

Name
ECS-Cluster-Node-1

Application and OS Images (Amazon Machine Image)
An AMI contains the operating system, application server, and applications for your instance. If you don't see a suitable AMI below, use the search field or choose **Browse more AMIs**.

Search our full catalog including 1000s of application and OS images

AMI from catalog Recents My AMIs Quick Start

Name
Bansir Amazon ECS Optimized Amazon Linux 2 v2023-0ea19bdd-8e7b-4450-aa67-403053969b11

Description
Bansir Amazon ECS Optimized Amazon Linux 2 v2023

Image ID
ami-0dbec2906e80dcef4

Username
root (Check with the AMI provider.)

Catalog	Published	Architecture	Virtualization	Root device type	ENA Enabled
AWS Marketplace AMIs	2023-09-13T13:17:59.000Z	x86_64	hvm	ebs	Yes

Summary
Number of instances: 1

Software Image (AMI)
Amazon ECS Optimized Amazon L...read more
ami-0dbec2906e80dcef4

Virtual server type (instance type)
t3.micro

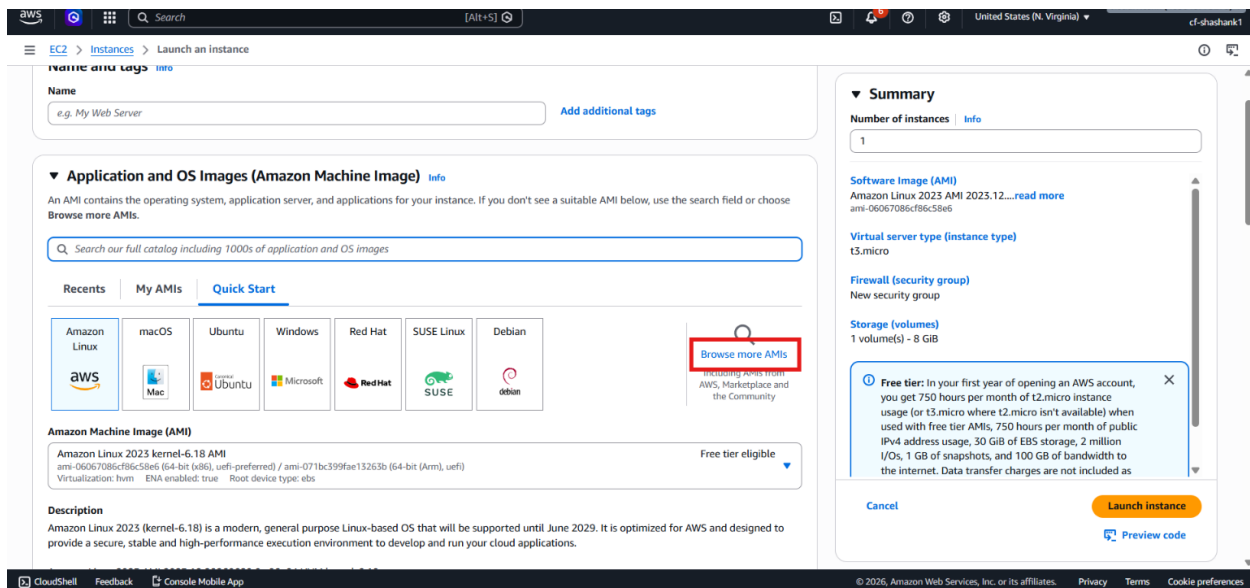
Firewall (security group)
New security group

Storage (volumes)
1 volume(s) - 30 GiB

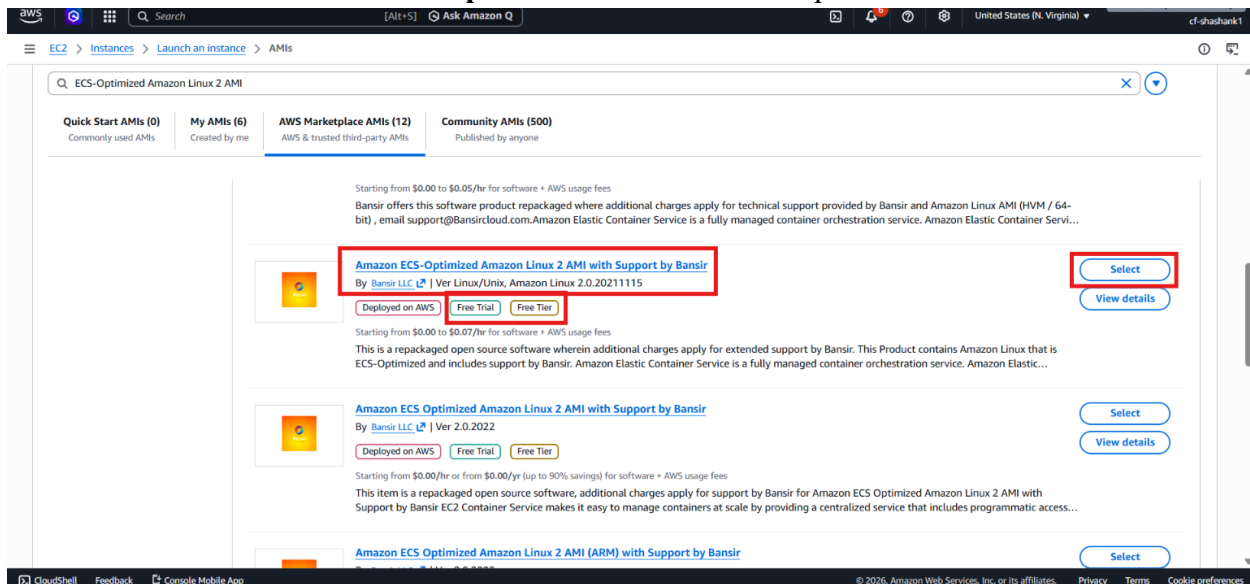
Free tier: In your first year of opening an AWS account, you get 750 hours per month of t3.micro instance usage (or t3.micro where t2.micro isn't available) when used with free tier AMIs, 750 hours per month of public IPv4 address usage, 30 GiB of EBS storage, 2 million I/Os, 1 GB of snapshots, and 100 GB of bandwidth to the internet. Data transfer charges are not included as part of the free tier allowance. Charges may apply depending on your account's free tier status.

Launch instance

- It can be found by simply searching and if not then follow these steps
 - Click on **Browse more AMIs**

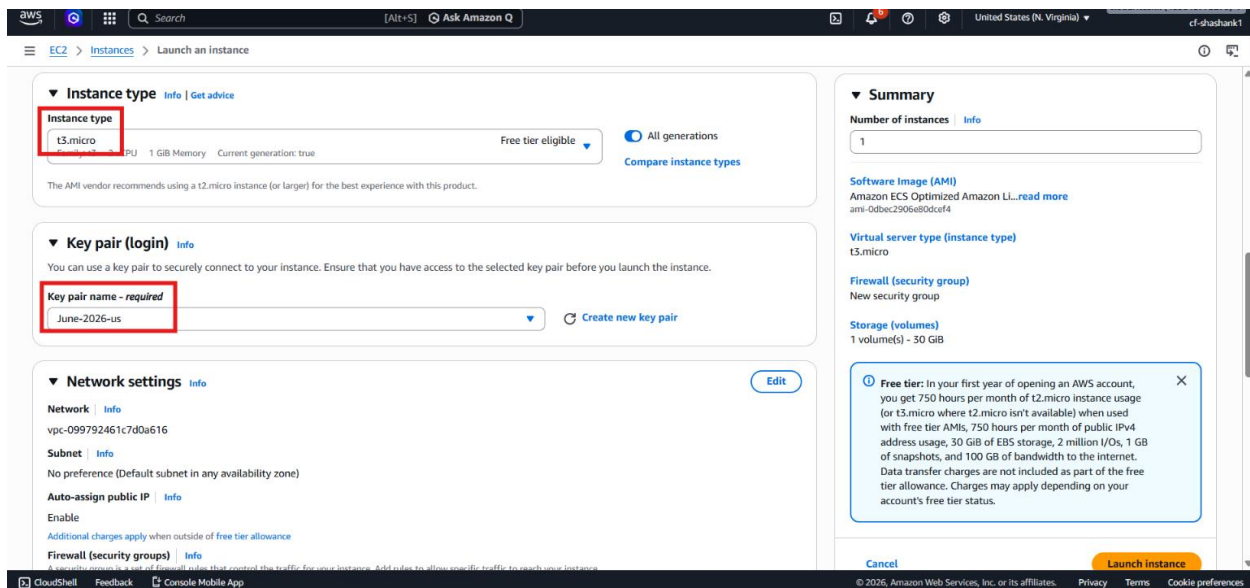


○ Select AWS Marketplace and search for the required AMI.



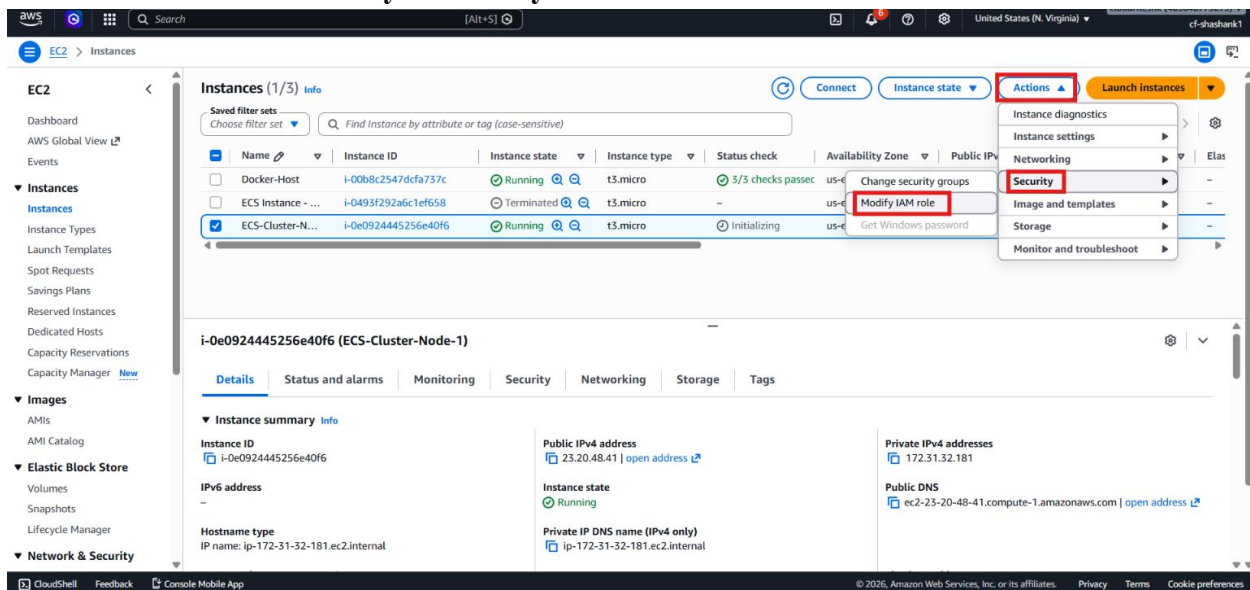
○ After selecting the required AMI, Click **Subscribe** and make sure it is of free tier otherwise you will be charged for the billing as per AMI.

- Use a free-tier instance type such as **t2.micro**.
- Select the correct VPC and subnet.
- Attach the key pair.
- Launch the instance.

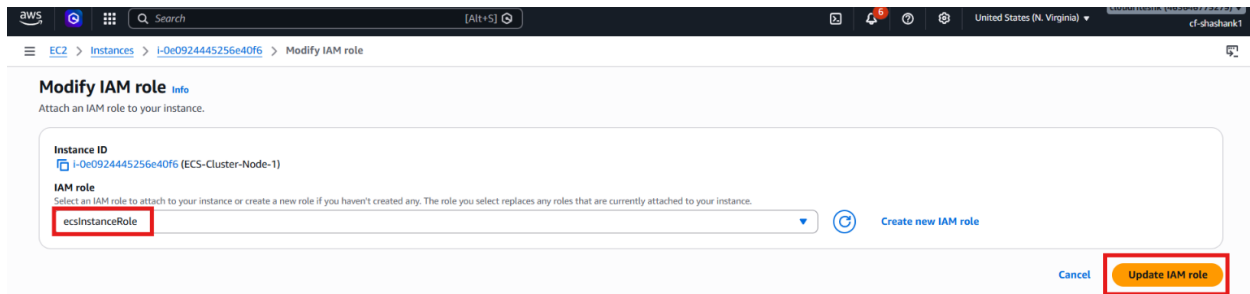


Step 3 – Attach the ECS Instance Role

- Open the EC2 instance.
- Go to **Actions** → **Security** → **Modify IAM role**.

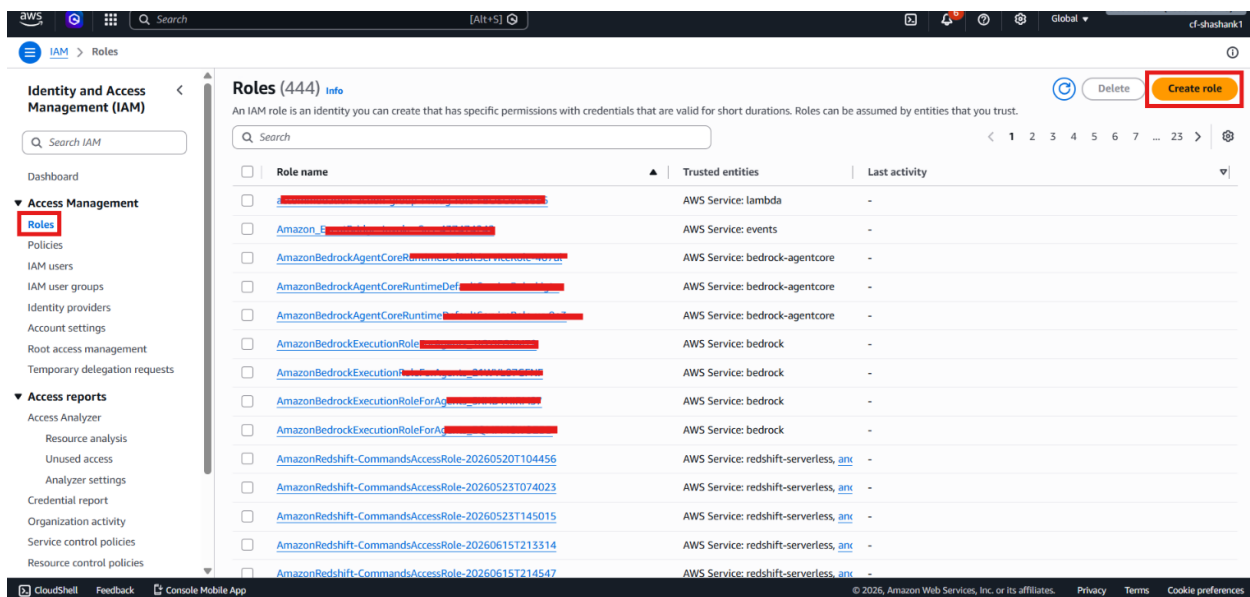


- Attach the role named: `ecsInstanceRole`



- If the role does not already exist:

- Open IAM
- Go to Roles
- Click **Create role**



- Select EC2

Trusted entity type

- ☒ **AWS service**
Allow AWS services like EC2, Lambda, or others to perform actions in this account.
- ☐ **AWS account**
Allow entities in other AWS accounts belonging to you or a 3rd party to perform actions in this account.
- ☐ **Web identity**
Allows users federated by the specified external web identity provider to assume this role to perform actions in this account.
- ☐ **SAML 2.0 federation**
Allow users federated with SAML 2.0 from a corporate directory to perform actions in this account.
- ☐ **Custom trust policy**
Create a custom trust policy to enable others to perform actions in this account.

Use case
Allow an AWS service like EC2, Lambda, or others to perform actions in this account.

Service or use case
EC2

Choose a use case for the specified service.

☒ **Use case**
EC2
Allows EC2 instances to call AWS services on your behalf.

- ☐ **EC2 Role for AWS Systems Manager**
Allows EC2 instances to call AWS services like CloudWatch and Systems Manager on your behalf.
- ☐ **EC2 Spot Fleet Role**
Allows EC2 Spot Fleet to request and terminate Spot Instances on your behalf.
- ☐ **EC2 - Spot Fleet Auto Scaling**
Allows Auto Scaling to access and update EC2 spot fleets on your behalf.
- ☐ **EC2 - Spot Fleet Termination**

○ Attach the policy: AmazonEC2ContainerServiceforEC2Role

Add permissions

Permissions policies (1/2326)

Choose one or more policies to attach to your new role.

Filter by Type: All types (1 match)

Policy name	Type	Description
☒ AmazonEC2ContainerServiceforEC2Role	AWS managed	Default policy for the Amazon EC2 Role f...

Set permissions boundary - optional

Cancel Previous **Next**

- Name the role: ecsInstanceRole
- Return to EC2 and attach it to the instance.

Step 4 – Configure the ECS Cluster Name on the EC2 Instance

- Connect to EC2 via SSH like we did earlier.


```
ec2-user@ip-172-31-32-181:~$ ssh -i "June-2026-us.pem" ec2-user@23.20.48.41
Microsoft Windows [Version 10.0.26200.8737]
(c) Microsoft Corporation. All rights reserved.

C:\Users\Shashank>cd downloads

C:\Users\Shashank\Downloads>ssh -i "June-2026-us.pem" ec2-user@23.20.48.41
The authenticity of host '23.20.48.41 (23.20.48.41)' can't be established.
ED25519 key fingerprint is SHA256:4x5+uKSm6gzQjQht+K6wxPYg9Ub0jLNuLHUFoXp4fo.
This host key is known by the following other names/addresses:
C:\Users\Shashank/.ssh/known_hosts:30: ec2-23-20-48-41.compute-1.amazonaws.com
Are you sure you want to continue connecting (yes/no/[fingerprint])? yes
Warning: Permanently added '23.20.48.41' (ED25519) to the list of known hosts.
Last login: Wed Sep 13 12:10:53 2023 from 195.181.167.197

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Amazon Linux 2 (ECS Optimized)

For documentation, visit http://aws.amazon.com/documentation/ecs
28 package(s) needed for security, out of 51 available
Run "sudo yum update" to apply all updates.
[ec2-user@ip-172-31-32-181 ~]$
```

- Open the ECS configuration file:

```
sudo vi /etc/ecs/ecs.config
```

```
ec2-user@ip-172-31-32-181:~$ ssh -i "June-2026-us.pem" ec2-user@23.20.48.41
(c) Microsoft Corporation. All rights reserved.

C:\Users\Shashank>cd downloads

C:\Users\Shashank\Downloads>ssh -i "June-2026-us.pem" ec2-user@23.20.48.41
The authenticity of host '23.20.48.41 (23.20.48.41)' can't be established.
ED25519 key fingerprint is SHA256:4x5+uKSm6gzQjQht+K6wxPYg9Ub0jLNuLHUFoXp4fo.
This host key is known by the following other names/addresses:
C:\Users\Shashank/.ssh/known_hosts:30: ec2-23-20-48-41.compute-1.amazonaws.com
Are you sure you want to continue connecting (yes/no/[fingerprint])? yes
Warning: Permanently added '23.20.48.41' (ED25519) to the list of known hosts.
Last login: Wed Sep 13 12:10:53 2023 from 195.181.167.197

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Amazon Linux 2 (ECS Optimized)

For documentation, visit http://aws.amazon.com/documentation/ecs
28 package(s) needed for security, out of 51 available
Run "sudo yum update" to apply all updates.
[ec2-user@ip-172-31-32-181 ~]$ sudo vi /etc/ecs/ecs.config
[ec2-user@ip-172-31-32-181 ~]$

[New] 1L, 31B written
```

- Add the ECS cluster name:

```
ECS_CLUSTER=your-cluster-name
```

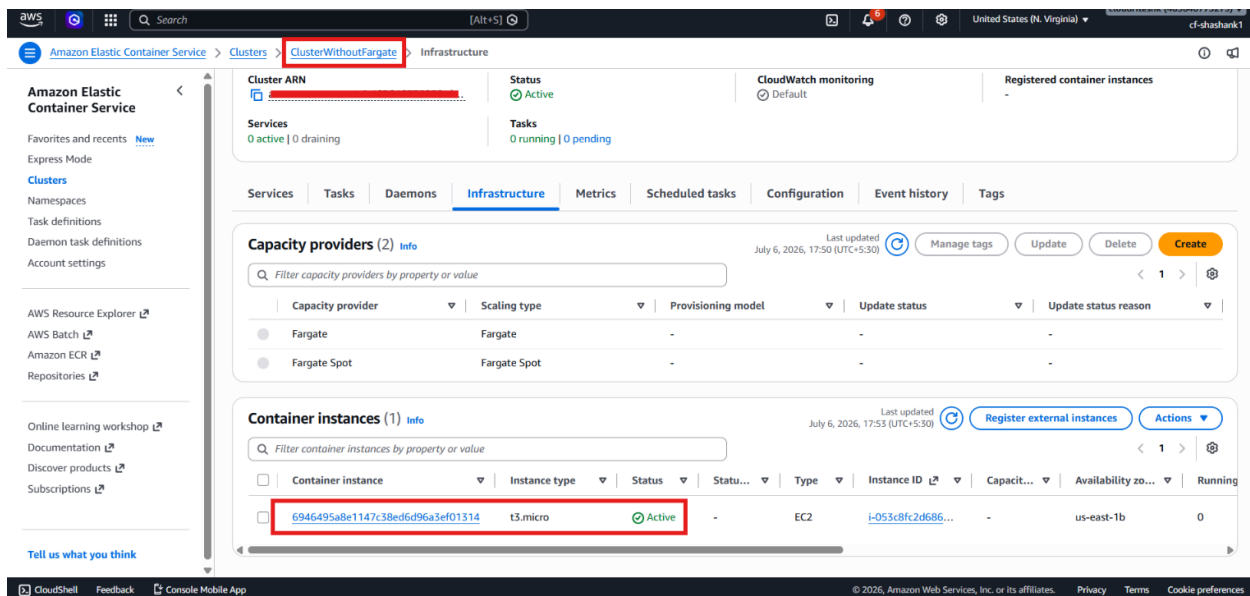
```
ec2-user@ip-172-31-32-181:~  
ECS_CLUSTER=ClusterWithoutFargate|  
  
-- INSERT --  
1,34 All
```

- Save the file.
- Restart the ECS service:

sudo systemctl restart ecs

```
ec2-user@ip-172-31-36-14:~  
C:\Users\Shashank>cd downloads  
C:\Users\Shashank\Downloads>ssh -i "June-2026-us.pem" ec2-user@34.207.249.254  
The authenticity of host '34.207.249.254 (34.207.249.254)' can't be established.  
ED25519 key fingerprint is SHA256:bQZqh9SARzhLgSsMlkmfMsRznvqhNhH1W4iDfxiUwo.  
This key is not known by any other names.  
Are you sure you want to continue connecting (yes/no/[fingerprint])? yes  
Warning: Permanently added '34.207.249.254' (ED25519) to the list of known hosts.  
Last login: Wed Sep 13 12:10:53 2023 from 195.181.167.197  
  
--|_--|_--|_--  
_|C|_|_|_|_ Amazon Linux 2 (ECS Optimized)  
----\_|_|_|_|_|  
  
For documentation, visit http://aws.amazon.com/documentation/ecs  
28 package(s) needed for security, out of 51 available  
Run "sudo yum update" to apply all updates.  
[ec2-user@ip-172-31-36-14 ~]$ sudo vi /etc/ecs/ecs.config  
[ec2-user@ip-172-31-36-14 ~]$ sudo systemctl restart ecs  
[ec2-user@ip-172-31-36-14 ~]$ sudo systemctl status ecs  
● ecs.service - Amazon Elastic Container Service - container agent  
   Loaded: loaded (/usr/lib/systemd/system/ecs.service; enabled; vendor preset: disabled)  
   Active: active (running) since Mon 2026-07-06 12:01:39 UTC; 9s ago  
     Docs: https://aws.amazon.com/documentation/ecs/  
   Process: 16010 ExecStopPost=/usr/libexec/amazon-ecs-init post-stop (code=exited, status=0/SUCCESS)  
   Process: 15996 ExecStop=/usr/libexec/amazon-ecs-init stop (code=exited, status=0/SUCCESS)  
   Process: 16053 ExecStartPre=/usr/libexec/amazon-ecs-init pre-start (code=exited, status=0/SUCCESS)  
   Process: 16031 ExecStartPre=/bin/bash -c if [ $(/usr/bin/systemctl is-active docker) != "active" ]; then exit 1; fi (code=exited, status=0/SUCCESS)  
   Main PID: 16094 (amazon-ecs-init)
```

- Wait for the instance to register with the ECS cluster.



Step 5 – Verify the EC2 Instance in ECS

- Return to the ECS cluster.
- Open the **Infrastructure** tab.
- Confirm that the EC2 instance now appears as a container instance.
- Verify that the instance ID matches the EC2 instance you launched.

Verification

- The ECS cluster was created without EC2 initially.
- An ECS-optimized EC2 instance was launched successfully.
- The ecsInstanceRole was attached to the instance.
- The ECS cluster name was added in /etc/ecs/ecs.config.
- The ECS service was restarted.
- The EC2 instance appeared in the ECS cluster infrastructure tab.